



ElectronicsTutorials

INDUCTORS AND INDUCTANCE

For Students, Professionals
and Beyond

eBook 6

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TABLE OF CONTENTS

1. Inductors and Inductance	1
2. Units of Inductance	2
3. Self-inductance of a Coil	2
4. Energy Stored in an Inductor	3
5. Connecting Inductors Together.	3
6. RL Charging Circuit	7
7. RL Discharging Circuit.	9
8. Inductors in AC Circuits.	9

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1. INDUCTORS AND INDUCTANCE

Inductors are simple passive two-terminal devices which have the ability to store energy in the form of a magnetic field within their coiled windings. Inductors basically consist of a single length of wire formed or wound into a coil. As such, they are designed to take advantage of the relationship that exists between magnetism and electricity when an electrical current flows through the coil winding.

A magnetic field is produced around any wire, cable, or conductor carrying an electric current and the higher the current, the higher the magnetic field produced around it. Also the nearer we get to the centre of the current carrying conductor, the stronger will be the magnetic effect. This then forms the basis of electro-magnetism.

So whenever an electrical current flows through a straight wire conductor, a magnetic flux is developed around it. If we take this length of wire and form it into a coil of N-turns, the magnetic flux surrounding the coil will become more concentrated around it. This has the effect of increasing its strength many times over compared with the magnetic flux produced by the original single straight length of wire.

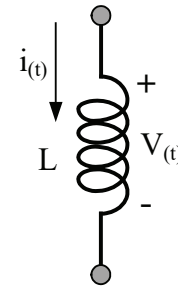
A magnetic field is created around a wire conductor by the electrical current flowing through it

However not all is that simple. As the strength of the magnetic flux increases, it creates a “self-induced” secondary voltage within the same coil, commonly called a “back emf” (electro-motive force) as a result of the primary magnetic field being produced.

Then for a coil of wire as shown in Figure 1. A self-induced voltage is developed across the coil due to the change in current flowing through the coil. The polarity of this self-induced voltage produces a secondary current in the coil that generates another magnetic flux which opposes any changes to the original flux.

In other words, the instant the main current begins to increase (or decrease) in value, there will be an opposing effect trying to limit this change. The amount of self-induced voltage opposing this change will depend on how fast the magnetic field is changing which itself is dependent on the frequency of the voltage supply.

FIGURE 1. VOLTAGE INDUCED IN A COIL



Because the length of wire in a coil is extremely long, the current through the coil cannot change instantaneously, but takes an amount of time for the current to change due mainly to the resistance and the self-induced effects of the wire coil. The ability of a coil to oppose any change in current is a result of its *self-inductance*, (**L**) properties.

The SI unit for self-inductance, simply called inductance, of an inductor is the Henry, named after Joseph Henry. It is defined as being: **“a coil will have an inductance value of one Henry when an emf of 1 volt is induced in the coil when the current flowing through the said coil changes at a rate of 1 ampere/second”**.

In other words, a coil has an inductance, (**L**) of one Henry, (1H) when the current flowing through it changes at a rate of one ampere/second, (A/s). This change induces a voltage of one volt, (**V_L**) in it. Thus the mathematical representation of the voltage induced in a coil with an inductance of **L** Henries as a result of this change in current is expressed as:

$$V_{(t)} = -L \frac{di_{(t)}}{dt} \text{ Volts}$$

Where: **di** is the change in the current in Amperes and **dt** is the time taken for this current to change in seconds. Note that the negative sign indicates that voltage induced opposes the change in current through the coil per unit time (*di/dt*).

If the inductor current is a constant steady state value, that is the current is not changing in magnitude, then the *di/dt* part of the equation becomes zero, which means zero back emf voltage is induced in the coil.

Since a current is flowing through the coil and zero voltage is developed across it, it must stand that the coil is now in a short circuit condition equal to a straight length of wire, or at the very least offering a very low resistive value, and this is true for an inductive coil used in DC circuits once the inductor current, **I_L** is constant.

In other words, the opposition to current flow offered by an inductor is very different between AC and DC circuits. Also note that inductance is the characteristic of a coil to oppose a change in current flow, whereas an inductor is a device which stores energy within itself in the form of a magnetic field. Thus inductance is the electrical characteristics of an inductor.

2. UNITS OF INDUCTANCE

Although the value of an inductor is measured in Henries, the Henry is a large unit of measurement so sub-multiples of the Henry are more commonly used instead. Inductance, L is always positive and has no negative units. For example.

Prefixes used for Henry (H)

- **Millihenry** (mH) $1\text{mH} = 1/1,000 = 0.001 = 10^{-3} \text{ H}$
- **Microhenry** (μH) $1\mu\text{H} = 1/1,000,000 = 0.000001 = 10^{-6} \text{ H}$

A typical inductor used in electrical and electronic circuits will have an inductance in the millihenry (mH) to microhenry (μH) range.

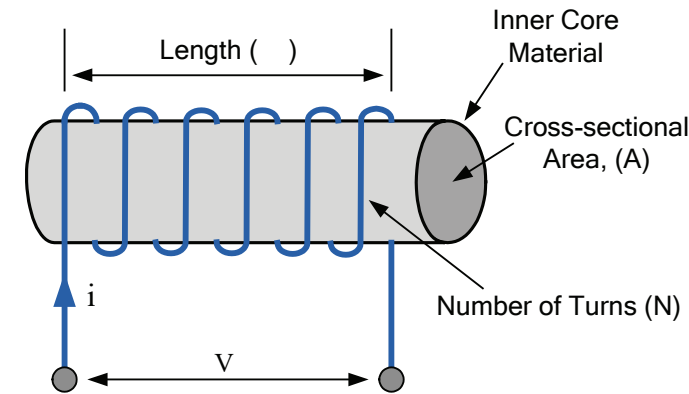
3. SELF-INDUCTANCE OF A COIL

Inductors are mainly designed to introduce specific amounts of inductance into a circuit. They are formed with wire tightly wrapped around a solid central core which can be either a straight cylindrical rod or a continuous loop or ring to concentrate their magnetic flux creating a much stronger magnetic field than one that would be produced by a simple coil of wire.

The inductance of a coil varies directly with the magnetic properties of the central core. Ferrite and powdered iron materials are mainly used for the core to increase the inductance by increasing the flux linking the coil.

Then the inductance of a coil which has a uniform cross-sectional area, A length, ℓ and coil turns, N can be represented in mathematical form as shown in Figure 2.

FIGURE 2. A SIMPLE INDUCTIVE COIL



$$L = \mu \frac{N^2 \times A}{\ell} \text{ Henries}$$

Where:

L = Inductance in Henries

N = The number of coil turns

μ = Permeability of the central core

A = Cross-sectional area in m^2

ℓ = Mean length of core in meters

From this we can see that the inductance of a coil depends on the physical shape of the coil, the type and permeability of the inner material, the number of turns and the length of the turns. So a coil of many turns will have a higher inductance value than one of only a few turns as more lines of magnetic flux exist to produce the required back emf.

The inductance of a coil can be increased further still if the coil is wound onto a ferromagnetic core, which is made of a soft iron material, such as soft iron, cobalt or nickel than a coil wound onto a non-ferromagnetic or hollow air core. This is because the material concentrates the lines of force more strongly through the softer ferromagnetic core material.

So for example, if the core material has a relative permeability 1000 times greater than free space, $1000\mu_0$ such as soft iron or steel, then the inductance of the coil would be 1000 times greater so we can say that the inductance of a coil increases proportionally as the permeability of the core increases.

4. ENERGY STORED IN AN INDUCTOR

When an electric current flows into an inductor, energy is stored in its magnetic field. If the current flowing through the inductor is increasing and di/dt becomes greater than zero, the instantaneous power in the circuit must also be greater than zero, ($P > 0$) ie, positive which means that energy is being stored in the inductor.

Likewise, if the current through the inductor is decreasing and di/dt is less than zero then the instantaneous power must also be less than zero, ($P < 0$) ie, negative which means that the inductor is returning energy back into the circuit.

As the power in an inductor is given as: $P = V_L \cdot I$, where as we have seen, $V_L = L di/dt$, the total magnetic energy which is always positive, being stored in the inductor is therefore given by the equation of Figure 3.

ENERGY STORED IN AN INDUCTANCE

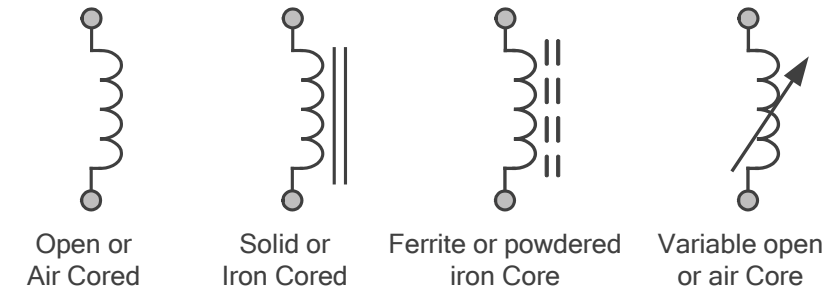
$$W_{(t)} = \frac{1}{2} L i_{(t)}^2 \text{ Joules}$$

Where: **W** is in joules, **L** is in Henries and **i** is the instantaneous current in Amperes

Note that the energy will remain stored in the inductors magnetic field as long as the current continues to flow, but as soon as the current falls or reduces to zero, the magnetic field reduces or collapses completely and the stored energy within the coil is returned back to the circuit. In an alternating current, AC circuit the inductor is constantly storing and delivering energy during each and every cycle.

The schematic symbol for an inductor is usually a single coil. Inductors are categorised according to the type of inner core they are wound around, for example, hollow core (free air), solid iron core or soft ferrite core with the different core types being distinguished by adding continuous or dotted parallel lines next to the wire coil.

FIGURE 3. INDUCTOR SYMBOLS



Inductors can be made so that their inductance is of a fixed type, for example 10mH or 100mH, or as a variable inductor by changing the relative positions of the core and coil as shown in Figure 3 for use in tv or radio tuning circuits. Generally, an arrow is used to indicate a variable inductor.

5. CONNECTING INDUCTORS TOGETHER

Inductors, just like resistors and capacitors, can be connected together in series or parallel combinations. Increasing levels of inductance can be obtained by connecting the inductors in series, while decreasing levels can be obtained by connecting inductors in parallel.

However, there are certain rules for connecting inductors in series or parallel and these are based on the fact that no mutual inductance or magnetic coupling exists between the individually connected inductors.

5.1 Inductors in Series

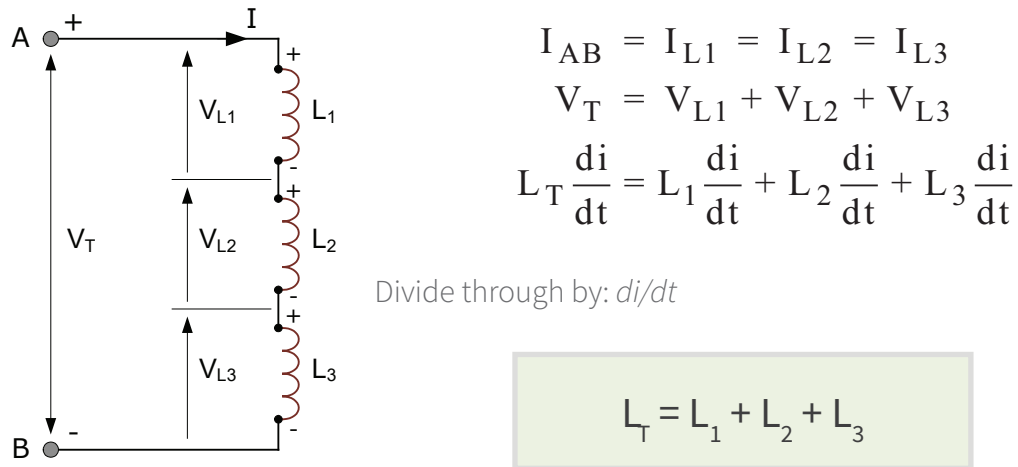
Inductors are said to be connected in “**Series**” when they are daisy chained together in a straight line, end to end. Inductors connected together in series are simply “added together” because the number of coil turns is effectively increased, with the total circuit inductance L_T being equal to the sum of all the individual inductances added together.

The current, (I_{AB}) that flows through the circuit of Figure 4, flows first through inductor, L_1 and then, as it has no other path to go, passes through the second inductor, L_2 and the third and so on as it has only one path to follow. Then, Inductors in Series all have a common current flowing through them. For example:

$$I_{L1} = I_{L2} = I_{L3} = I_{AB} \dots \text{etc.}$$

Figure 4 shows three inductors connected together in series between points A and B.

FIGURE 4. INDUCTORS IN SERIES



Then the total inductance of the series chain shown in figure 4 can be found by simply adding together the individual inductances of the inductors in series. One important point to remember about inductors in series circuits, the total inductance (L_T) of any two or more inductors connected together in series will always be GREATER than the value of the largest inductor in the series chain.

5.2 Mutually Connected Inductors in Series

When a current flows through a coil, a magnetic field is created around it. If we now place a second coil next to the first, when the current in the first coil changes, the varying flux

affects the second coil located nearby, thus inducing voltages in both as the magnetic field of one links with the other.

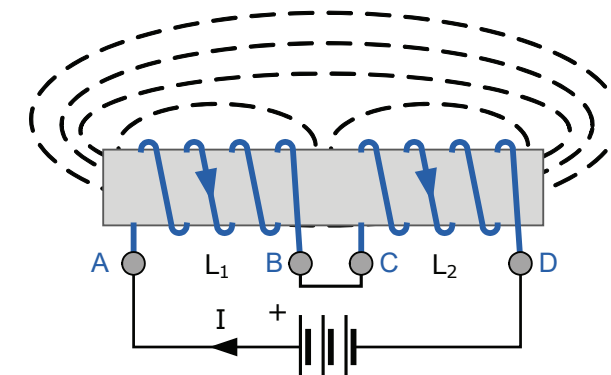
Therefore, any change of current in one coil will have an effect on the other adjacent coil through the production of mutually induced emf's and vice versa. The action of one coil being magnetically-coupled and inducing an emf into other nearby coils is called Mutual Induction, (M).

The effect of mutual inductance either increases or decreases the total inductance depending upon the amount of magnetic coupling and the distance apart of the coils as well as their orientation to each other. The unit of mutual inductance is still the henry, with the symbol, L_M .

Mutual Inductance is the effect that one coil has on another coil close by

Mutually connected series inductors can be classed as either "Aiding" or "Opposing" the total inductance. If the magnetic flux produced by the current flows through the coils in the same direction, then the coils are said to be Cumulatively Coupled. Likewise, if the current flows through the coils in an opposite direction, then the coils are said to be Differentially Coupled.

FIGURE 5. MUTUALLY CONNECTED SERIES INDUCTORS



In Figure 5 the two coils are cumulatively coupled as they are both wound in the same direction on the core and the same current flows through both coils in the same direction.

While the current flowing between points A and D through the two series-connected cumulatively coupled coils is in the same direction, the previous equation for the voltage drops across each of the coils needs to be modified to take into account the interaction between the two coils due to the effect of mutual inductance.

The self-inductance of each individual coil, L_1 and L_2 respectively will be the same as before but with the addition of L_M denoting the mutual inductance. Then the total emf induced into the cumulatively coupled coils is given as:

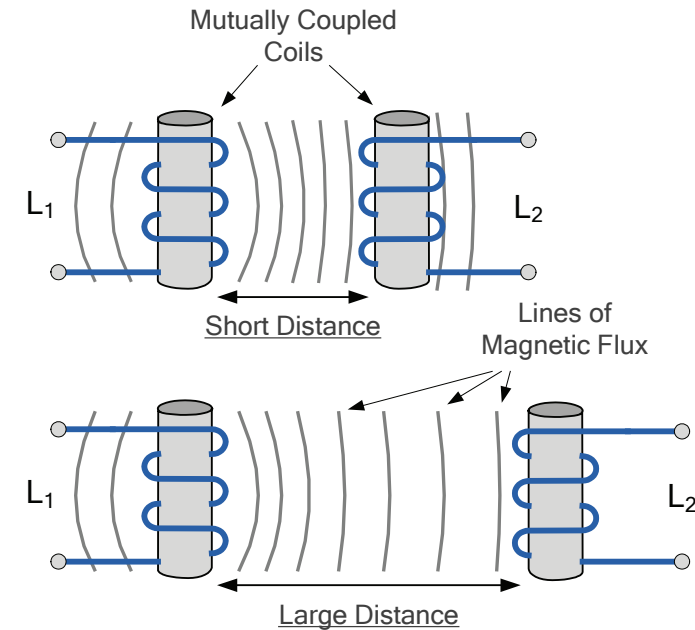
$$L_T = L_1 + L_2 \pm 2L_M$$

Where L_M is the mutual inductance between the coils with the plus (+) sign being used if the two coils are connected in series-aiding configuration, while the minus (-) sign is used if the two coils are connected in a series-opposing configuration.

However, the amount of mutual inductance that links one coil to another depends very much on the relative positioning of the two coils. If one coil is positioned next to the other coil so that their physical distance apart is small, then nearly all of the magnetic flux generated by the first coil will interact with the coil turns of the second coil inducing a relatively large emf and therefore producing a large L_M mutual inductance value.

Likewise, if the two coils are more apart from each other or at different relative angles, the amount of induced magnetic flux from the first coil into the second would be a lot weaker producing a much smaller induced emf and therefore a much smaller mutual inductance value. So the effect of mutual inductance is very much dependent upon the relative positions or spacing between the coils and this is demonstrated in Figure 6.

FIGURE 6. MUTUAL INDUCTANCE BETWEEN TWO COILS



Mutual inductance signifies some form of magnetic coupling exists between two inductors or coils, depending on their distance and arrangement. The self inductance of each individual coil is given as:

$$L_1 = \mu \frac{N_1^2 \times A}{\ell} \quad \text{and} \quad L_2 = \mu \frac{N_2^2 \times A}{\ell}$$

By cross-multiplying the two equations above, the mutual inductance, L_M that exists between the two coils can be expressed in terms of the self inductance of each coil.

$$L_M = \sqrt{L_1 L_2} \quad \text{Henries}$$

However, the equation assumes zero flux leakage and 100% magnetic coupling between the two coils, L_1 and L_2 .

In reality there will always be some loss due to leakage and position, thus the magnetic coupling between two coils can never reach or exceed 100%, but can come very close to this value in some special inductive coils.

The amount of mutual inductance between the two coils is defined as being the ratio of the flux that links the two coils to the total flux produced by the energising coil given as a percentage fraction. This fractional value is called the coefficient of coupling and is given the letter “**k**”.

$$k = \frac{\text{The mutual flux linking } L_1 \text{ and } L_2}{\text{Total flux produced by } L_1}$$

The amount of inductive coupling that exists between the two coils is generally expressed as a fractional number between 0 and 1 instead of a percentage (%) value, where 0 (zero) indicates no inductive coupling, and 1 indicating full or maximum inductive coupling.

In other words, if $k = 1$ the two coils are perfectly coupled, if $k > 0.5$ the two coils are said to be tightly coupled and if $k < 0.5$ the two coils are said to be loosely coupled. Mutual inductance increases with tighter coupling and therefore a greater coefficient of coupling value. Then the previous equation which assumed a perfect coupling can be modified to take into account of this coefficient of coupling, k and is given as:

$$L_M = k\sqrt{L_1 L_2} \quad \text{or} \quad k = \frac{L_M}{\sqrt{L_1 L_2}}$$

Note that k has no units because it is a ratio of two values of magnetic flux.

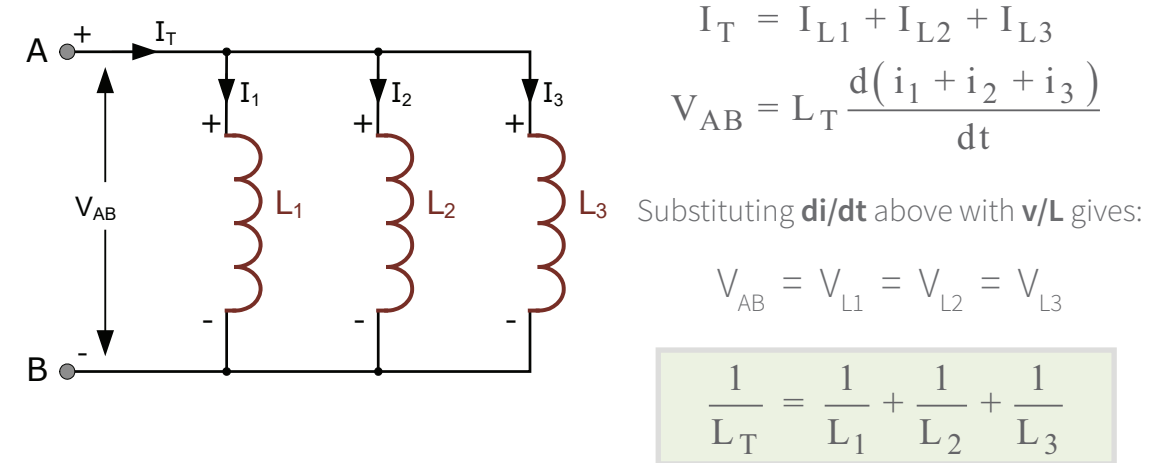
5.3 Inductors in Parallel

Inductors are said to be connected together in “**Parallel**” when both of their terminals are respectively connected to each terminal of the other inductor or inductors. The voltage drop across all of the inductors in parallel will be the same. Then, Inductors in Parallel have a Common Voltage across them and is given as:

$$V_{AC} = V_{L1} = V_{L2} = V_{L3} \dots \text{etc.}$$

Figure 7 shows three inductors connected together in parallel between points A and B.

FIGURE 7. INDUCTORS IN PARALLEL



Thus if “ n ” inductances are connected together in parallel, the reciprocal of the total inductance, L_T is equal to the sum of the reciprocals of the individual inductances. One important point to remember about inductors in parallel circuits, the total inductance (L_T) of any two or more inductors connected together in parallel will always be LESS than the value of the smallest inductance in the parallel chain.

Note that the above equation for inductors in parallel only holds true when there is “NO” mutual inductance or magnetic coupling between two or more of the inductors, (they are magnetically isolated from each other). Where there is coupling between coils, the total inductance is also affected by the amount of coupling.

5.4 Mutually Connected Inductors in Parallel

As with series connected inductors, parallel connected inductors are also influence by mutual inductance with the effect of this mutual inductance depending upon the distance apart of the coils and their orientation to each other.

Mutually connected inductors in parallel can be classed as either “aiding” or “opposing” the total inductance depending on the direction the currents are flowing through each coil. Parallel aiding connected coils increase the total equivalent inductance, while parallel opposing coils decrease the total equivalent inductance compared to coils which have zero mutual inductance coupling.

Mutual coupled parallel coils can be shown as either connected in an aiding or opposing configuration by the use of polarity dots or polarity markers. Figure 8 shows parallel aiding inductors in which both polarity dots are connected to the same terminal, and Figure 9 shows parallel opposing inductors as both polarity dots are connected to opposite terminals.

FIGURE 8. PARALLEL AIDING COILS

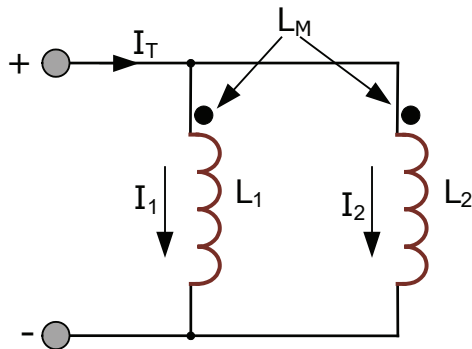
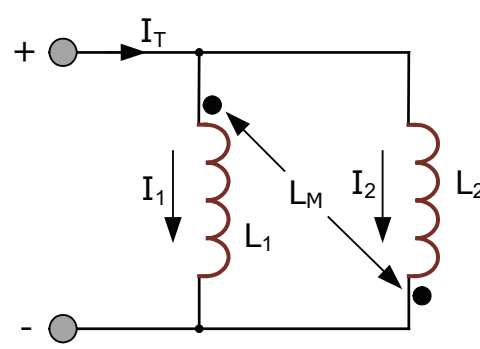


FIGURE 9. PARALLEL OPPOSING COILS



For two parallel inductors with mutual coupling, the standard formula for the equivalent inductance in Henries is:

$$L_T = \frac{L_1 L_2 - L_M^2}{L_1 + L_2 \pm 2L_M} \text{ Henries}$$

If the two coils are in an aiding configuration, then the self and mutually induced emf's are equal so the $-L_M$ part of the equation is used. If one coil is reversed with respect to the other, we would then have two parallel opposing coils so the mutually induced emf's will have a cancelling effect on each other so the $+2L_M$ part of the equation is used.

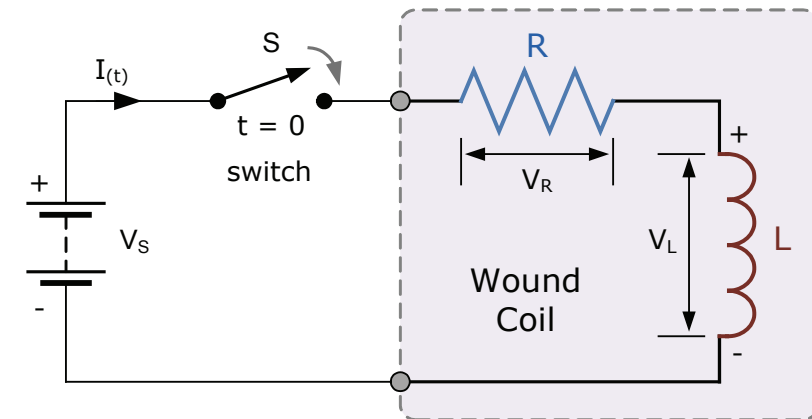
6. RL CHARGING CIRCUIT

A Resistor and an Inductor can be connected together in series to form an RL circuit. Resistor-inductor (RL) circuits are widely used in a variety of applications such as electronic filters and any application requiring the use of an electromagnetic field.

Thus far we have examined the inductor as a pure or ideal passive component but in the real world “ALL” coils whether they are inductors, chokes, solenoids, relays or any wound component will always have a certain amount of resistance no matter how small. This is because the actual coils turns of wire being used to make it uses copper wire which has a resistive value.

An RL Series Circuit consists basically of an inductor of inductance, L connected in series with a resistor of resistance, R . The resistance “ R ” is generally the DC resistive value of the copper wire turns or loops that goes into making up the inductors coil. Consider the RL series circuit shown in Figure 10.

FIGURE 10. AN RL CHARGING CIRCUIT



The RL series circuit of Figure 10 is connected to a constant voltage source, V_S and an ON/OFF switch, S . Assume that switch S is open until it is closed at a time $t = 0$, and then remains permanently closed producing a “step response” type voltage input.

With the switch closed, current (I) begins to flow around the circuit but does not rise rapidly to its maximum value of I_{MAX} as determined by the ratio of V / R (Ohms Law) because, as we have seen previously, a back emf is induced into the coil windings as a result of the rate of change of current flowing through it.

Thus Kirchhoff's voltage law (KVL) gives us: $V_{S(t)} = V_R + V_L$

As the voltage across the resistance is equal to $I \times R$, and the voltage across the inductor is now our familiar expression of $L(di/dt)$. The final expression for the individual voltage drops around the RL series circuit can be given as:

$$V_{S(t)} = I \times R + L \frac{di}{dt} \quad \text{Volts}$$

As the voltage drop across the resistor depends upon the current, i , while the voltage drop across the inductor depends upon the rate of change of the current, di/dt . When the current is equal to zero, ($i = 0$) at time $t = 0$ the above expression, which is also a first order differential equation, can be rewritten to give the value of the charging current at any instant of time as:

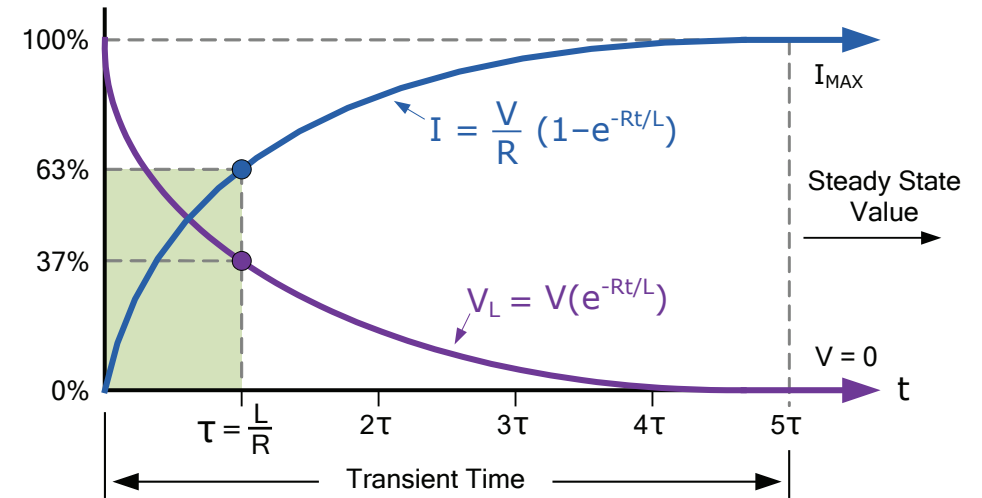
$$i(t) = \frac{V_S}{R} \left[1 - e^{-Rt/L} \right]$$

Where: V = Volts, R = Ohms, L = Henries, t = Seconds, and e is the base of Natural Logs.

The Time Constant, (τ) of the RL series circuit is given as L/R . Current increases from 0 to V_S/R and once it reaches this maximum steady state value at 5τ , the back emf, V_L across the inductor is zero because a constant current is now flowing in the circuit with the inductor acting more like a short circuit and effectively removing it from the circuit.

Therefore, the current flowing through the coil is limited only by the resistive element in Ohms of the coils windings. A graphical representation of the current growth representing the voltage/time characteristics of the circuit can be presented in Figure 11 as.

FIGURE 11. RL VOLTAGE AND CURRENT CHARGING CURVES



Since the voltage drop across the resistor, V_R is equal to $I \times R$ (Ohms Law), it will have the same exponential growth and shape as the current. However, the voltage given across the inductor, V_L will have a value equal to: $V(e^{-Rt/L})$.

The voltage across the inductor, V_L will have an initial value equal to the supply voltage V_S at time $t = 0$, or when the switch is first closed, and then decays exponentially to zero.

The voltage and current in an energising and de-energising RL circuit takes 63% of change during each time constant

The inductor voltage drops to e^{-1} ($\approx 37\%$) of its initial value within one time constant, tau (τ) as represented by the characteristics curves of Figure 11.

The time required for the current flowing in the RL series circuit to reach its maximum steady state value is equivalent to about 5 time constants or 5τ . This time constant τ , is measured by $\tau = L/R$, in seconds, where R is the value of the resistor in ohms and L is the value of the inductor in Henries.

7. RL DISCHARGING CIRCUIT

If an inductor has been connected to a supply voltage for a period greater than 5 time constants, 5τ or $5RC$, the circuit current will be at its maximum steady state value. The energy used to achieve this is stored within the magnetic field produced around the inductor's coil as a result of the flow of the steady state current.

When the voltage supply is removed and replaced by a short circuit, the magnetic flux created around the inductive coil collapses because there is no supply voltage to maintain the flow of current through the inductor.

Thus the magnetic field around the inductor begins to slowly decay as the inductor attempts to discharge itself back through the series resistor, R in the reverse direction.

However at the same time, the steady state current flowing through the inductive coil begins to reduce to zero, which by its very action creates a change of flux around the coil as it is no longer in a steady state condition. This action self-induces a voltage, V_L into the same coil which tries to oppose and slow down the rate of decay of the current.

The polarity of this self-induced voltage across the inductor is in the reverse direction compared to the charging direction of the inductor. As there is still a current flowing through the inductor, but with a magnitude decaying towards zero, the value of the discharging current at any instant of time can be given as:

$$i(t) = \frac{V_L}{R} \left[\epsilon^{-Rt/L} \right] \text{ Amperes}$$

Where: V_L/R is the steady state current flowing in the inductor at the instant in time the coil was shorted, starting at time, $t = 0$.

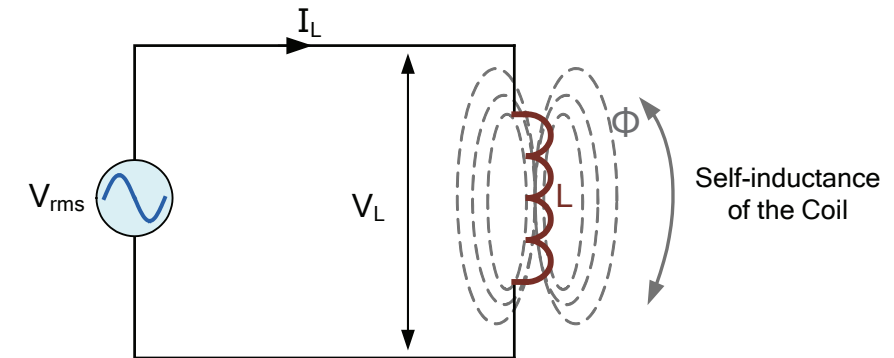
Then we can see that an RL circuit discharges (or charges) as an exponential function of time because the current is restricted to not changing instantaneously, unlike the voltage across the inductor.

8. INDUCTORS IN AC CIRCUITS

We have seen that inductors store energy within their coil turns in the form of a magnetic flux. When an inductor is connected across a DC supply voltage the rate of change of current through its coil creates a back emf which opposes the increasing flux at a rate determined by its L/R time constant, and the faster the rate of change in current the greater the back emf.

However, when connected to an alternating sinusoidal supply voltage, an inductor will alternately charge and discharge at a rate determined by the frequency of the supply. Thus for inductors operating in a time-varying (AC) circuit, their magnetic field is constantly reversing every half cycle. Consider the inductive circuit of Figure 12.

FIGURE 12. INDUCTOR WITH A SINUSOIDAL SUPPLY



In the purely inductive circuit of Figure 12, the inductor is connected directly across the AC supply voltage. As the supply voltage increases and decreases with the frequency, the self-induced back emf also increases and decreases in the coil with respect to this change opposing current flow through it.

The opposition to current flowing through a coil in an AC circuit is determined by its AC resistance, more commonly known as Impedance (Z). But resistance is always associated with DC circuits so to distinguish AC resistance from DC resistance the term Reactance is generally used.

Just like resistance, the value of reactance is also measured in Ohm's but is given the symbol **X**, (uppercase letter "X"), to distinguish it from a purely resistive value. As the component we are interested in is an inductor, the reactance of an inductor is therefore called "**Inductive Reactance**".

In other words, an inductor's electrical resistance when used in an AC circuit is called *Inductive Reactance* because it is the "reaction" of the inductor to the changing value of alternating current.

The opposition to current flowing in an AC inductor is known as Inductive Reactance, X_L

Inductive reactance, symbol X_L , is the property in an AC circuit which opposes the change in the current through an inductor. The opposition to current flow being proportional to both the amount of inductance and to the frequency of the supply. Thus, increasing supply frequency increases the inductive reactance, and as frequency increases, the rate of change of current increases, increasing the value of the induced voltage.

The formula used for Inductive Reactance multiplies the frequency of the AC wave by the value of Inductance:

Inductive Reactance Formula:

$$X_L = 2\pi fL$$

Where:

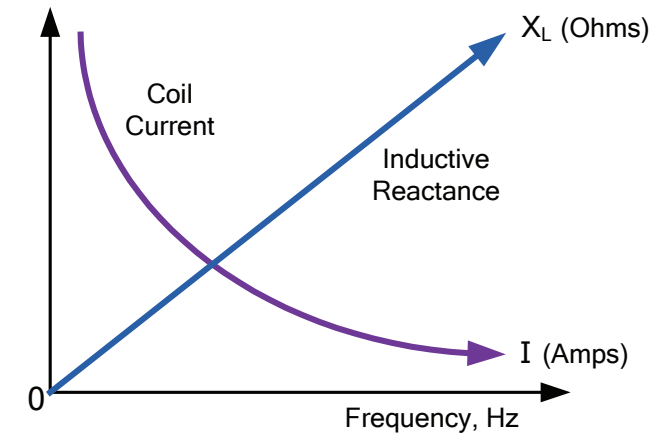
X_L = Inductive Reactance (Ohms), f = frequency (Hertz) and L = inductance (Henry).

From the above equation for inductive reactance, it can be seen that if either of the Frequency or Inductance is increased, the overall inductive reactance value would also increase proportionally. As the frequency approaches infinity the inductor's reactance would also increase to infinity with the inductor acting like an open circuit condition.

However, as the frequency approaches zero or DC, the inductor's reactance would decrease to zero, acting like a short circuit. This means then that inductive reactance is directly "proportional" to frequency.

In other words, inductive reactance increases with frequency resulting in X_L being small at low frequencies and X_L being high at high frequencies and this is demonstrated in the graph of Figure 13.

FIGURE 13. INDUCTIVE REACTANCE AGAINST FREQUENCY



The slope of Figure 13 shows that the "Inductive Reactance" of an inductor increases as the supply frequency across it increases.

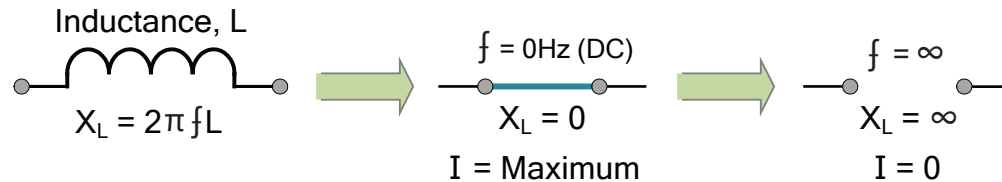
Therefore the inductive reactance of an inductor increases as the frequency across it increases therefore inductive reactance is proportional to frequency ($X_L \propto f$) as the back emf generated in the inductor is equal to its inductance multiplied by the rate of change of current in the inductor.

Thus in an AC circuit, inductive reactance, X_L depends on the frequency of the supply voltage and the amount of inductance of the inductor.

Also as the frequency increases the current flowing through the inductor also reduces in value. Then we can see that at DC (0Hz) an inductor has zero reactance (short-circuit), at very high frequencies (infinity) an inductor will have infinite reactance (open-circuit).

We can present the effect of very low and very high frequencies on the reactance of a pure AC Inductance as shown in Figure 14.

FIGURE 14. EFFECT OF FREQUENCY ON A PURE INDUCTOR



Thus, at high frequencies an inductor acts like an open circuit, and at low frequencies or DC voltages, an inductor acts like a short circuit (piece of wire).

8.1 The Frequency Domain

In an AC circuit containing a pure inductance, the maximum current flowing through the inductor is given as:

$$I_{MAX} = \frac{V_{MAX}}{X_L} \quad \text{where: } X_L = 2\pi fL$$

and therefore, the rms current flowing through an AC inductance will be defined as:

$$I_{L(t)} = I_{MAX} \sin(\omega t - 90^\circ)$$

Where: $I_L = V/2\pi fL$ (or $I_L = V/X_L$) is the current magnitude and $\theta = -90^\circ$ is the phase angle or difference between the voltage and current. For a purely inductive circuit, I_L lags V_L by 90° (ELI), or V_L leads I_L by 90° .

8.2 The Phasor Domain

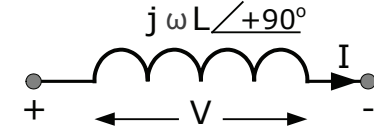
In the phasor domain the voltage across the plates of an AC inductance will be:

$$V_L = j\omega L \times I_L$$

$$\text{where: } j\omega L = jX_L = \text{Impeadance (Z)}$$

and in Polar Form this would be written as: $X_L \angle +90^\circ$ where:

$$X_L \angle \theta = \frac{V_L \angle +90^\circ}{I_L \angle 0^\circ}$$



and:

$$X_L \angle \theta = j\omega L = 0 + jX_L = \omega L \angle +90^\circ = Z \angle +90^\circ$$

End of this Inductors eBook

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With the completion of this Inductors and Inductance eBook you should have gained a basic understanding and knowledge of inductors. The information provided here should give you a firm foundation for continuing your study of electronics and electrical engineering. In ebook 7 we will learn about AC Circuit Theory.

For more information about any of the topics covered here please visit our website at:

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